

Technologies and Tools Used

This project, titled “**A Comprehensive Measure of Well-Being**”, was developed using a robust set of Python-based tools and libraries. These technologies enabled efficient data processing, machine learning model development, visualization, and deployment of the final solution.

1. Development Environment

Anaconda Navigator

Anaconda Navigator was used to manage Python environments and install required libraries. It provides a user-friendly interface for organizing packages and launching development tools.
<https://www.anaconda.com/download>

PyCharm IDE

PyCharm was used as the primary development environment for writing, testing, and debugging the project code. It offers advanced features such as code completion and error detection, improving development efficiency.
<https://www.jetbrains.com/pycharm/>

2. Data Processing and Analysis

NumPy

NumPy was used for numerical computations and handling multi-dimensional arrays, which are essential for efficient data processing.

Pandas

Pandas was utilized for data cleaning, transformation, and analysis. It helped in organizing the dataset into structured formats like DataFrames for better manipulation.

3. Machine Learning

Scikit-learn

Scikit-learn was used to build and evaluate machine learning models. Algorithms such as classification and regression were applied to analyze well-being indicators and generate meaningful insights.

4. Data Visualization

Matplotlib

Matplotlib was used to create basic plots and charts for data visualization.

Seaborn

Seaborn enhanced the visual representation by providing more attractive and informative statistical graphs, helping in better understanding of patterns and trends.

5. Deployment

Flask

Flask was used to develop a lightweight web application for deploying the machine learning model. It allows users to interact with the model through a simple interface.

Conclusion

The combination of these tools ensured smooth development, accurate analysis, and effective deployment of the project. Together, they contributed to building a reliable system for measuring and analyzing well-being.